

1. Introduction & Background

1.1 Problem Statement

Sierra Leone is ranked among the countries with the lowest access to safe drinking water in West Africa. According to the World Health Organization (WHO), less than 60% of Sierra Leone's population has access to basic water services, and only 15% have access to safely managed drinking water. Contaminated water sources, broken infrastructure, pipeline leakages, and inadequate sanitation facilities disproportionately affect low-income urban communities and rural villages.

The existing water issue reporting system relies on manual, paper-based processes that are slow, prone to data loss, and disconnected from response teams. Community members have no direct channel to report water problems to authorities, and WASH officers lack real-time data to prioritize and coordinate responses efficiently.

1.2 Project Purpose

AquaGuard SL was conceived as a technology-driven solution to this problem. The system creates a direct digital channel between citizens, local water authorities, NGOs, and health workers, enabling rapid issue reporting, transparent tracking, and coordinated resolution.

1.3 Objectives

- Enable community members to report water issues in real time using mobile devices
- Provide GPS and photo-verified evidence to support issue validation
- Give WASH officers and local authorities a dashboard to track, verify, and resolve issues
- Alert affected communities about water contamination and service disruptions
- Generate analytics and reports to guide evidence-based water infrastructure investment
- Align with UN SDG 6 (Clean Water and Sanitation) as a Digital Public Good

1.4 Target Users

- Community Users (Citizens/Residents) – Mobile app users who report issues and receive alerts
- Local Authority / WASH Officers – Web portal users who verify and resolve issues
- NGO / Health Workers – Web portal users who monitor health impacts and coordinate responses
- System Administrators – Technical staff who manage the platform, users, and data

2. System Overview

2.1 About AquaGuard SL

AquaGuard SL (Sierra Leone) is a web and mobile application platform that digitises the water issue reporting and management process. The name 'AquaGuard' reflects the system's core mission: to guard and protect water resources and the communities that depend on them.

The platform operates on a three-tier architecture, separating the user interface (Presentation Layer), business logic (Application Layer), and data storage (Database Layer). This architecture ensures scalability, security, and maintainability across the system's lifetime.

2.2 Key Features

- Real-time water issue reporting with GPS location tagging and photo uploads
- Interactive water map showing issue hotspots, water sources, and resolution status
- Automated alert and notification system for contamination warnings and issue updates
- Role-based access control for citizens, WASH officers, NGOs, and administrators
- Issue tracking dashboard with Pending → In Progress → Resolved lifecycle management
- Community feedback and rating system for service quality measurement
- Analytics dashboard with charts, heatmaps, and exportable reports
- Response coordination tools for NGOs and government agencies

3. Software Development Life Cycle (SDLC)

3.1 SDLC Model Selected: Agile (Scrum Framework)

AquaGuard SL is developed using the Agile SDLC model, specifically the Scrum framework. Agile was chosen because the project requirements are subject to change based on stakeholder feedback, and the system needs to be delivered iteratively with continuous improvement.

Agile development allows the team to work in 2-week sprints, delivering functional increments of the system at each sprint review. This approach aligns with the project's community-driven nature — regular feedback from community users and WASH officers shapes each iteration of the product.

3.2 Why Agile Over Other Models?

Model	Limitation	Agile Advantage
Waterfall	Rigid; cannot accommodate changing requirements	Flexible; adapts to stakeholder feedback at each sprint
Spiral	High cost and complexity; best for large enterprises	Cost-effective for small teams with iterative delivery
V-Model	Testing only at the end; late bug discovery	Continuous testing integrated into every sprint

3.3 SDLC Phases for AquaGuard SL

#	Phase	AquaGuard SL Activities
1	Planning	Identified the water quality problem in Sierra Leone. Set project scope, team roles, timeline (Gantt chart), and budget. Aligned project with UN SDG 6 (Clean Water and Sanitation).
2	Requirements Analysis	Gathered functional and non-functional requirements through stakeholder interviews (community members, WASH officers, NGOs). Produced a Software Requirements Specification (SRS).
3	System Design	Designed the 3-layer architecture (Presentation, Application, Database). Created UML diagrams: Use Case, Class, Sequence, Activity, ER Diagram. Designed UI wireframes in Figma.
4	Implementation	Prototype developed in Figma (high-fidelity). Codebase organised on GitHub. Tech stack: React Native (mobile), Node.js (backend), PostgreSQL (database), Google Maps API (water map).
5	Testing	Unit testing, integration testing, and UAT (User Acceptance Testing). Feedback collected from 20 test users. Bug tracking via GitHub Issues.
6	Deployment	System deployed on cloud infrastructure (AWS). Mobile app published on Google Play Store and Apple App Store. Handover to local authority WASH department.
7	Maintenance	Monthly updates, performance monitoring, security patches, user feedback collection, and iterative feature improvements based on community usage data.

3.4 Agile Sprint Plan

AquaGuard SL is developed across 9 active working weeks (Weeks 1–11, with Week 10 as buffer and Week 11 as submission). The following sprint breakdown guides the development:

Sprint	Weeks	Deliverables
Sprint 1	Weeks 4–	Project planning, requirements gathering, stakeholder analysis, SRS draft
Sprint 2	Weeks 5 - 6	System architecture, UML diagrams (Use Case, Class, Sequence, ER), GitHub setup
Sprint 3	Weeks 7–8	Figma UI/UX prototype (10+ screens), design review and revision
Sprint 4	Weeks 9–10	Full documentation draft, Gantt chart, Network diagram, GitHub commits
Sprint 5	Weeks 11	Testing, review, final report formatting, GitHub finalisation, submission

4. System Requirements

4.1 Stakeholder Analysis

The following stakeholders were identified and consulted during the requirements gathering phase of AquaGuard SL:

- Community Users (Citizens/Residents) — Primary users who experience water issues daily
- Local Authority WASH Officers — Government water and sanitation personnel
- NGO / Health Workers — Non-governmental organisations monitoring public health impacts
- System Administrators — Technical team responsible for platform management
- Ministry of Water Resources — Government oversight and policy alignment

4.2 Functional & Non-Functional Requirements

ID	Requirement	Description	Priority
FR-01	User Registration & Login	Users shall register using email/phone and log in securely.	High
FR-02	Issue Reporting	Community users can report water issues with GPS and photo evidence.	High
FR-03	Water Map	Interactive map showing water sources, issue hotspots, and real-time markers.	High
FR-04	Alert & Notification	System sends alerts for new issues, contamination warnings, and resolution updates.	High
FR-05	Issue Verification	WASH officers verify reported issues before escalation.	Medium
FR-06	Issue Resolution	Authorities update issue status (Pending → In Progress → Resolved).	High
FR-07	Feedback & Rating	Users rate service quality after issue resolution.	Medium
FR-08	Analytics Dashboard	Admins view reports, heatmaps, and trend data.	Medium
FR-09	Response Coordination	NGOs and authorities coordinate response efforts in-platform.	Medium
FR-10	My Reports	Users view their submitted reports and track status.	High

4.3 Use Case Summary

The Use Case Diagram (presented in Section 6.1) captures the interactions between actors and the system. Key use cases include:

- Community User: Register/Login, Report Issue (with GPS and Photo), View Water Map, Receive Alerts, Give Feedback
- WASH Officer: Verify Issue, Resolve Issue, View Reports
- NGO/Health Worker: Access Analytics, Coordinate Response
- System Administrator: Manage Users, Manage System Data

5. System Architecture Design

5.1 Architecture Overview

AquaGuard SL follows a Three-Tier Architecture pattern, which separates the application into three logical layers: the Presentation Layer (User Interface), the Application Layer (Business Logic), and the Database Layer (Data Storage). This separation of concerns ensures that each layer can be maintained, scaled, and updated independently.

5.2 System Architecture Diagram

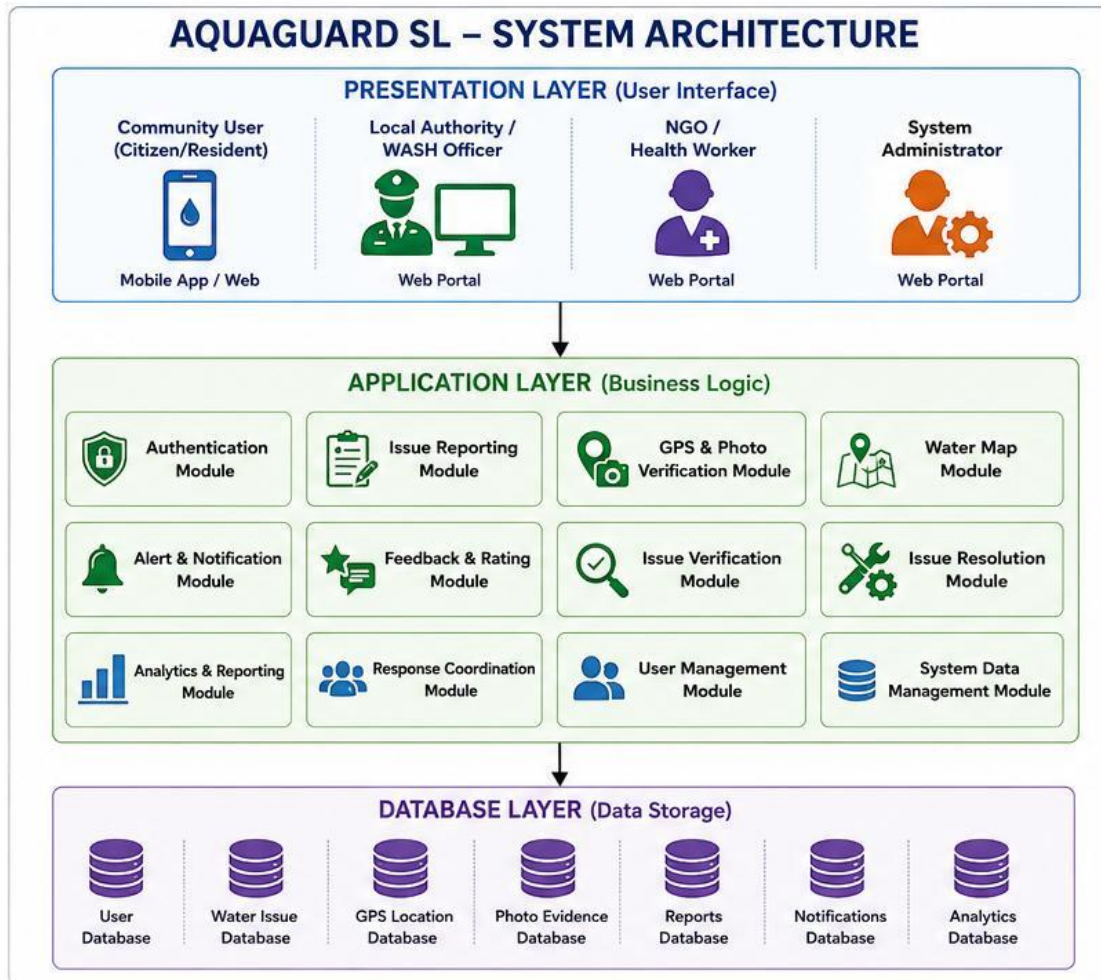


Figure 1: AquaGuard SL – Three-Tier System Architecture

5.3 Layer Descriptions

5.3.1 Presentation Layer (User Interface)

The Presentation Layer is the front-end interface through which different user types interact with AquaGuard SL. Four distinct user interfaces are provided:

- **Community User (Mobile App / Web)**: A mobile-first interface allowing citizens to report issues, view the water map, receive alerts, track their reports, and provide feedback. Built with React Native for cross-platform iOS and Android support.

- Local Authority / WASH Officer (Web Portal): A web dashboard for verifying reported issues, updating resolution status, viewing reports, and managing field teams.
- NGO / Health Worker (Web Portal): A monitoring portal for accessing analytics, viewing affected areas, and coordinating response activities.
- System Administrator (Web Portal): A full-access administrative panel for managing users, roles, system data, and platform configurations.

5.3.2 Application Layer (Business Logic)

The Application Layer contains twelve functional modules that process all business rules and operations. Each module is independently deployable as a microservice:

Module	Description
Authentication Module	Handles user registration, login, password reset, and session management.
Issue Reporting Module	Allows community users to submit water issues with category selection, description, GPS coordinates, and photo uploads.
GPS & Photo Verification Module	Validates that reported GPS coordinates are real locations and compresses/stores photo evidence securely.
Water Map Module	Renders an interactive map powered by Google Maps API showing water sources, issue markers, and status overlays.
Alert & Notification Module	Sends push notifications, SMS, and in-app alerts for contamination warnings, issue updates, and area advisories.
Feedback & Rating Module	Collects user satisfaction ratings (1-5 stars) after issue resolution to measure service quality.
Issue Verification Module	Enables WASH officers to review submitted reports, request additional evidence, and verify authenticity.
Issue Resolution Module	Tracks issue lifecycle (Pending → In Progress → Resolved) and notifies stakeholders at each transition.
Analytics & Reporting Module	Provides dashboards with charts, heatmaps, trend analysis, and exportable PDF/CSV reports for administrators.
Response Coordination Module	Facilitates communication between NGOs, health workers, and authorities to coordinate response actions.
User Management Module	Admins manage user accounts, roles, permissions, and access levels across the platform.

Module	Description
System Data Management Module	Handles database backups, data archiving, log management, and system configuration settings.

5.3.3 Database Layer (Data Storage)

The Database Layer consists of seven specialised databases hosted on AWS RDS (PostgreSQL), each serving a distinct data domain:

- User Database — Stores user profiles, credentials, roles, and authentication tokens
- Water Issue Database — Stores all reported issues with status, category, and metadata
- GPS Location Database — Stores GPS coordinates linked to each reported issue
- Photo Evidence Database — References to photo files stored in AWS S3
- Reports Database — Stores generated analytics reports and historical data
- Notifications Database — Stores notification history and delivery status
- Analytics Database — Aggregated data for dashboards, heatmaps, and trend analysis

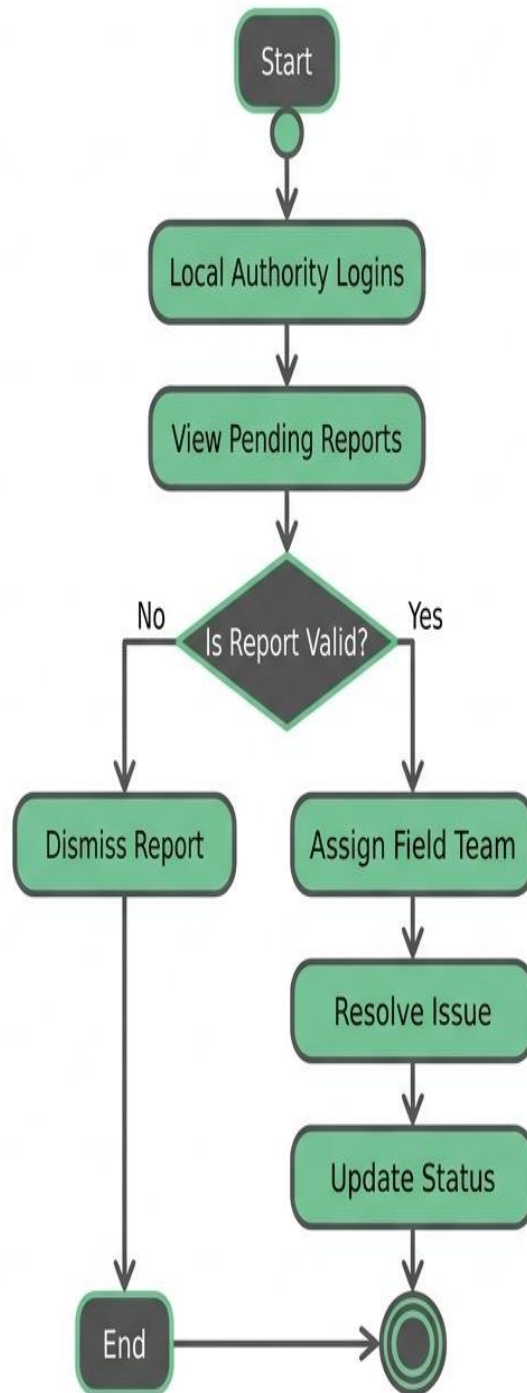
5.4 Component Interaction

Data flows through the system as follows: A community user submits a water issue report through the mobile app (Presentation Layer). The Issue Reporting Module in the Application Layer validates the input, invokes the GPS & Photo Verification Module to authenticate location and image data, and stores the report in the Water Issue Database and GPS Location Database. The Alert & Notification Module then triggers push notifications to nearby users and WASH officers. When an officer updates the issue status, the Issue Resolution Module updates the Water Issue Database and sends a resolution notification back to the original reporter.

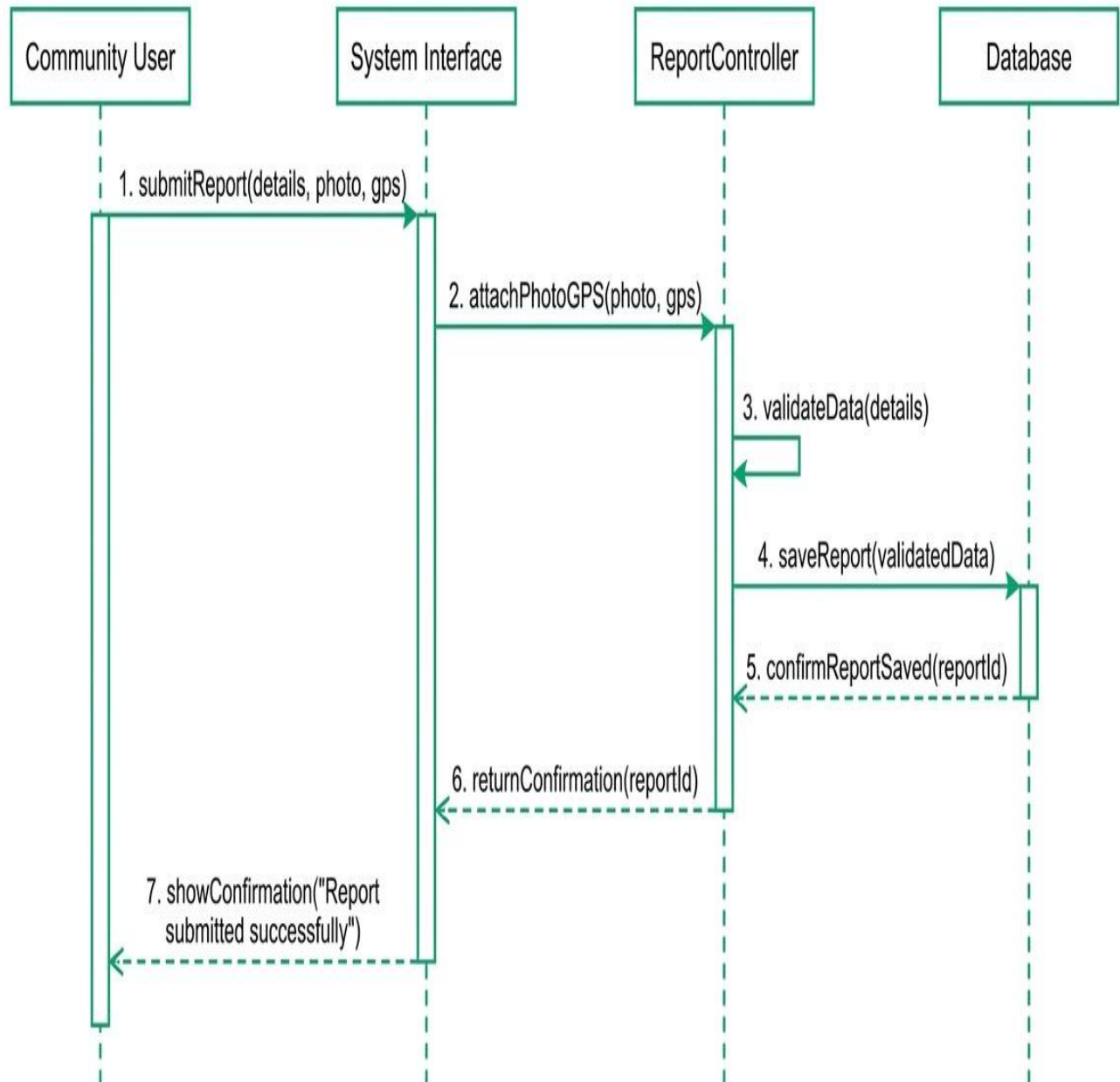
All inter-layer communication uses secure RESTful APIs over HTTPS. The Application Layer never directly exposes database connections to the Presentation Layer, ensuring a secure and maintainable architecture.

6. UML Diagrams

AquaGuard SL - Issue Verification and Resolution Process



UML Sequence Diagram: "Report Issue" Flow - AquaGuard SL System



Emerald Green: Interaction/Lifeline
Charcoal Grey: Text/Labels

6.1 Use Case Diagram

The Use Case Diagram illustrates the interactions between the four system actors (Community User, Local Authority/WASH Officer, NGO/Health Worker, and System Administrator) and the system's use cases. All use cases include a Login dependency (shown as include relationships), ensuring that only authenticated users can access system functions.

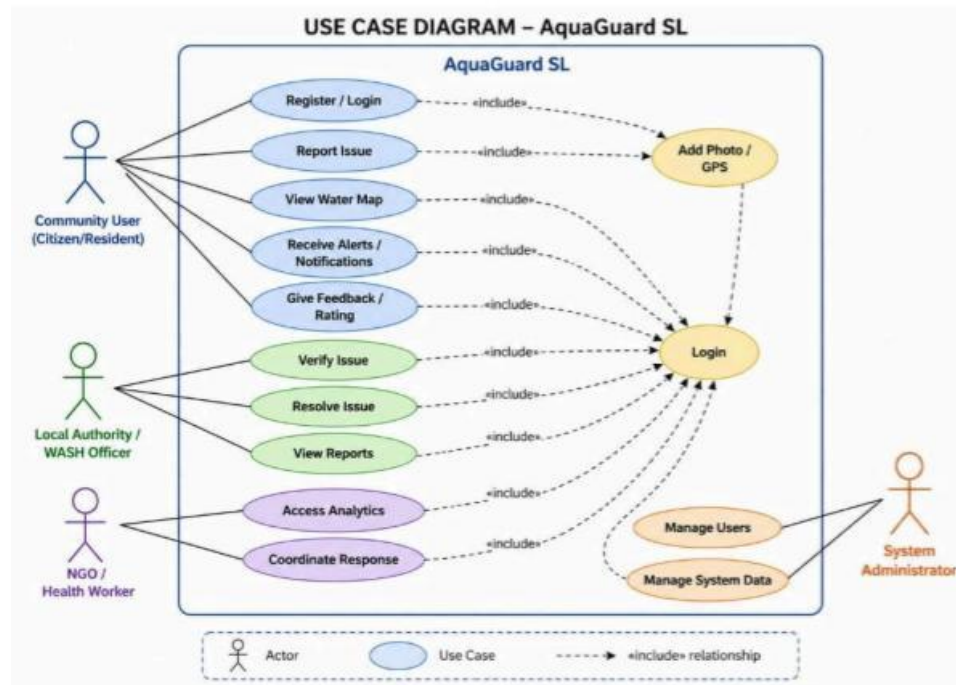


Figure 2: AquaGuard SL – Use Case Diagram

7. Prototype Development (UI/UX Design)

7.1 Design Tools

The AquaGuard SL prototype was designed using Figma, the industry-standard collaborative design tool. Figma was selected for its real-time collaboration features, component libraries, and the ability to create high-fidelity interactive prototypes that closely simulate the final application experience.

7.2 Design Principles

- Simplicity — Clean, uncluttered layouts accessible to users with low digital literacy
- Consistency — Uniform use of the brand colour palette (blue/purple #3D2BFF primary, white background) across all screens
- Accessibility — Large touch targets, readable font sizes (minimum 14pt), and high contrast ratios
- Mobile-First — Designed for 375x812px (iPhone 17) base dimensions, adaptable to Android

- Feedback — Every user action provides immediate visual or notification feedback

7.4 Screen Designs

The following high-fidelity screens were designed in Figma and represent the core user journeys of AquaGuard SL. The prototype covers all required user roles and key use cases identified in the Use Case Diagram.



4:50



Welcome Back!

Login to continue



Email or Phone

Enter email or phone

Password

Enter password



☒ Remember me

[Forget Password?](#)

Login

Don't have an account? [Register](#)

4:50



Create Account



Join AquaGuard

Full Name

Email

Phone Number

Password

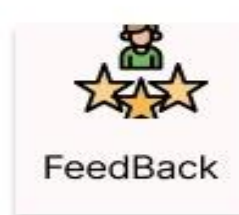
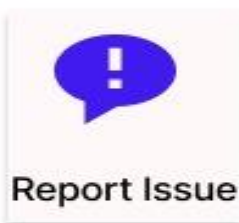
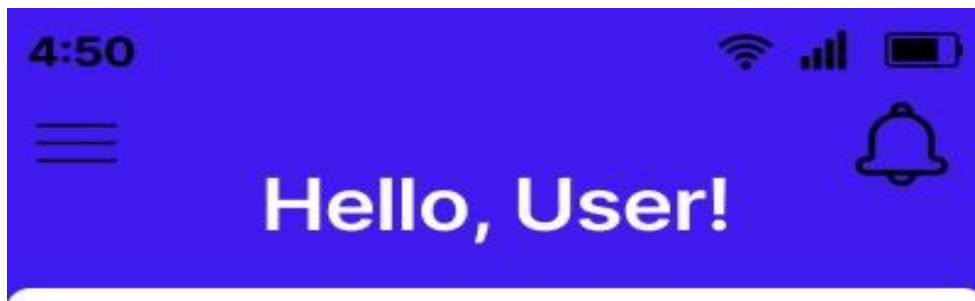


Confirm Password



Register

Already have an account? [Login](#)



4:50



Report New Issue

Issue Type

select issue type



Location

Tap to get current location



Description

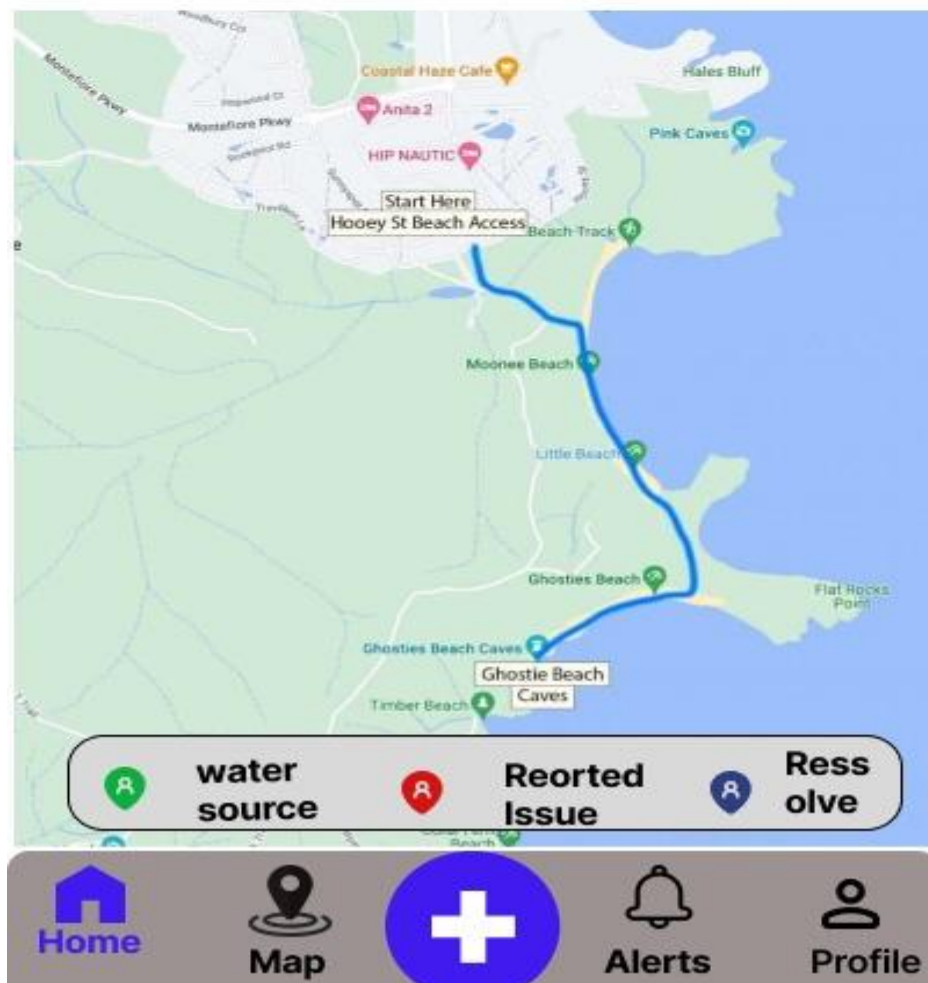
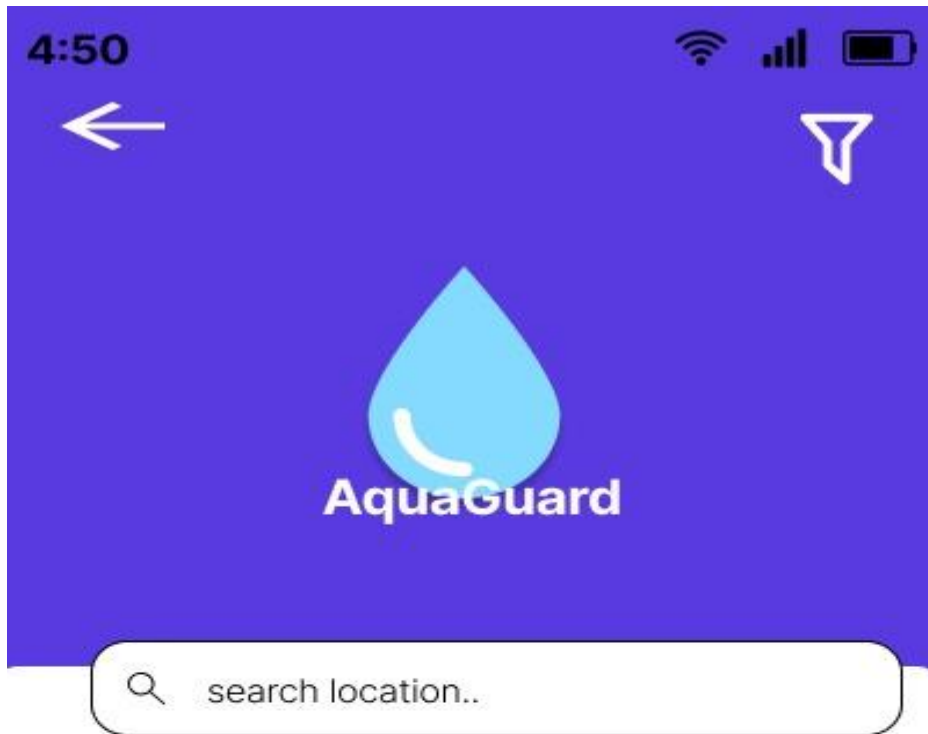
Describe the issue...

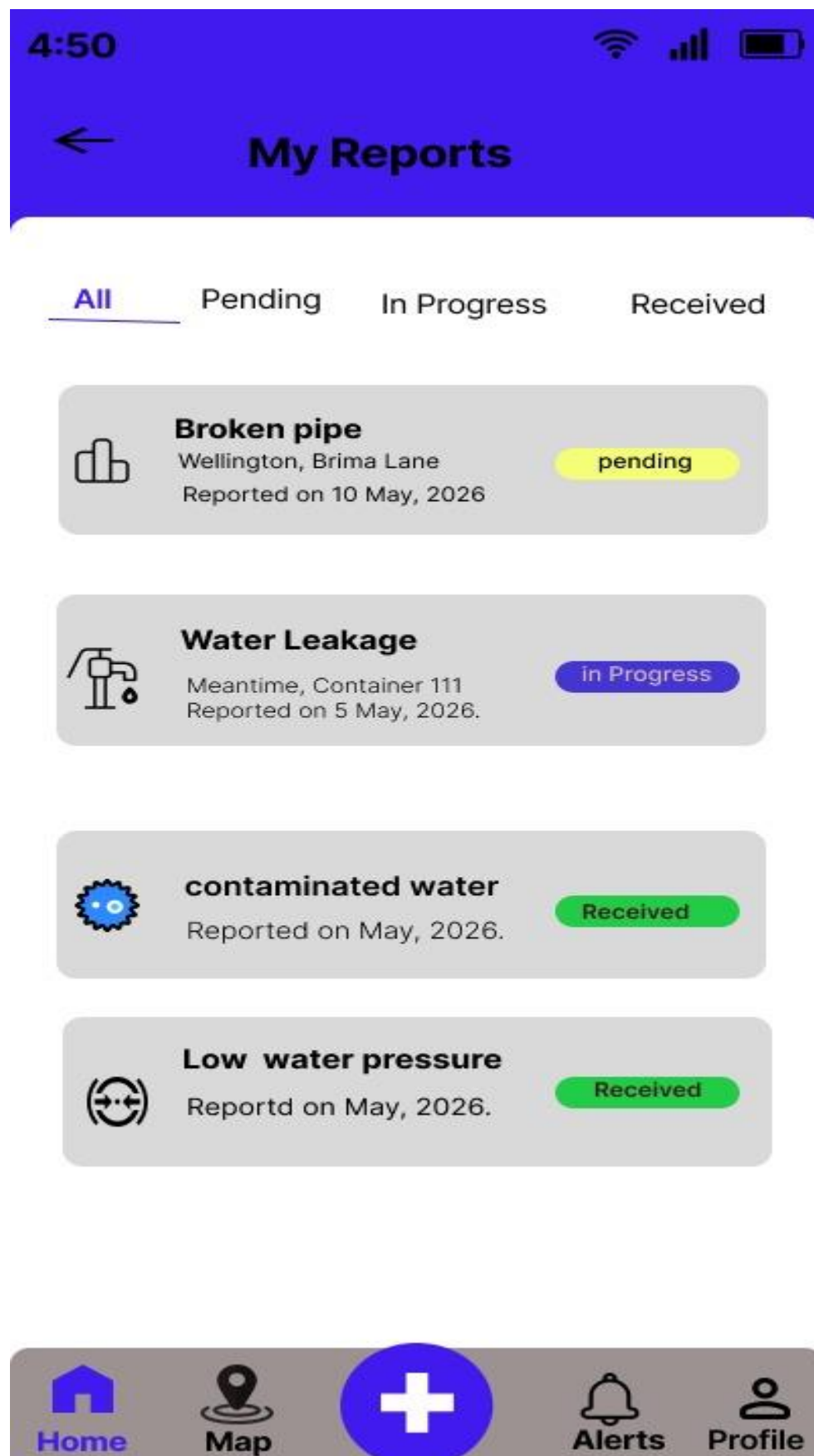
Upload Photo

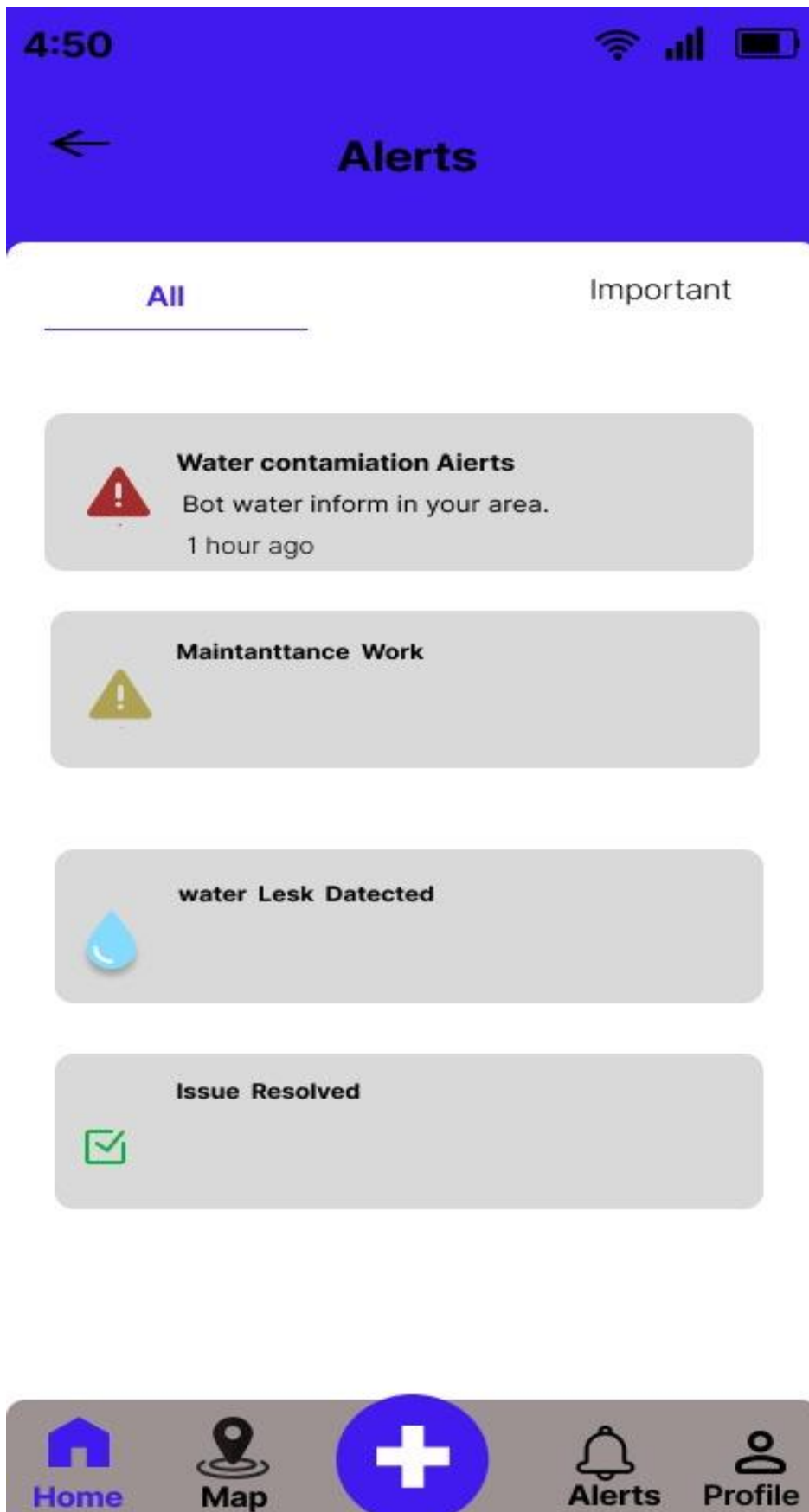


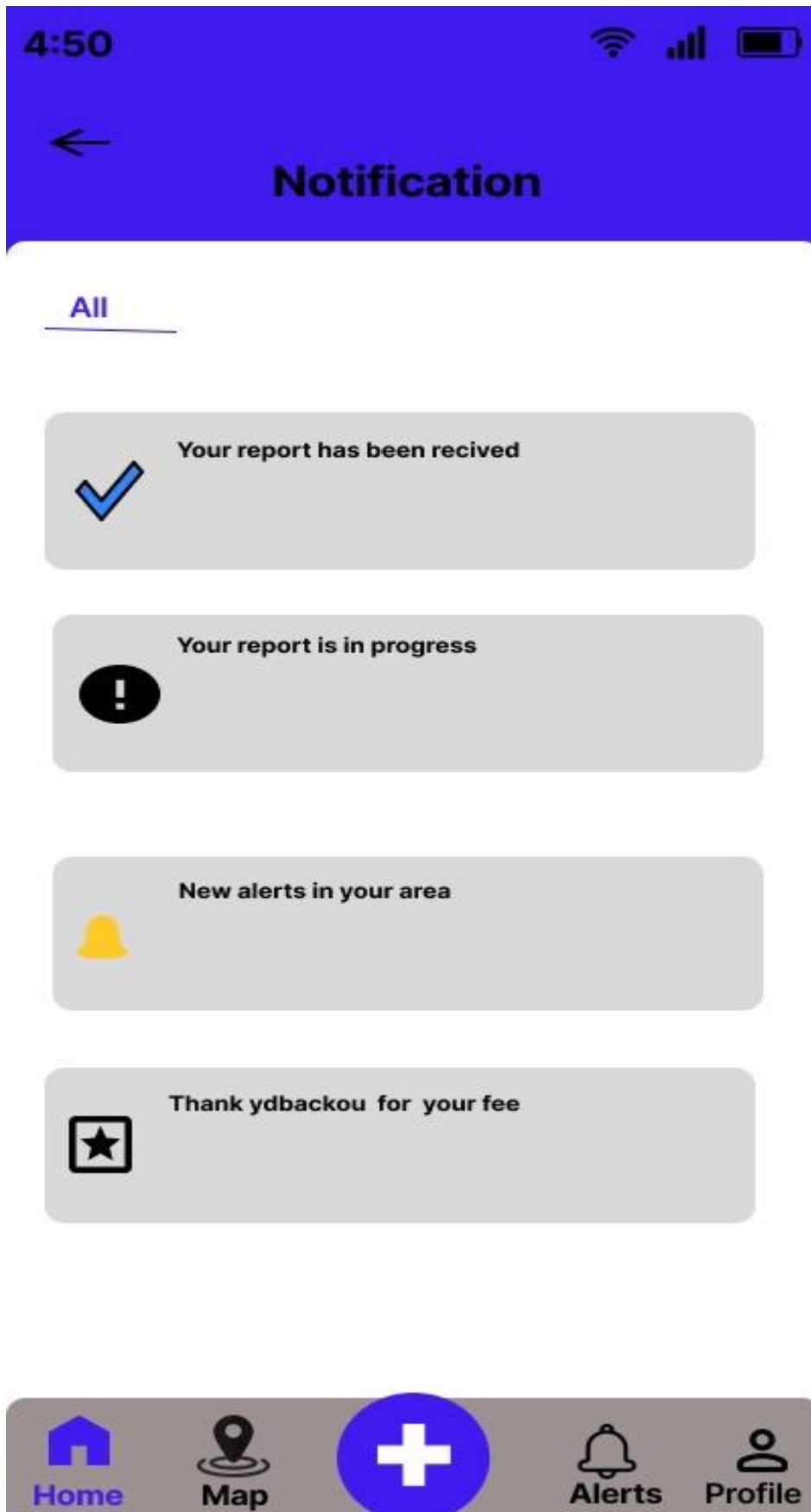
Add Photo

Submit Report









8. Git & GitHub Integration

8.1 Repository Overview

Item	Details
Repository Name	aquaguard-sl
GitHub URL	https://github.com/aquaguard-sl/aquaguard-sl
License	MIT License
Default Branch	main
Branches	main, develop, feature/ui, feature/backend, feature/docs
Key Files	README.md, /docs (report), /design (Figma exports), /diagrams (UML)
Collaborators	All team members added as contributors
Issues Tracker	Used for bug tracking and task management
README	Includes project overview, setup guide, SDG alignment, and license

8.2 Repository Structure

The AquaGuard SL GitHub repository is organised as follows:

- / (root) — README.md, LICENSE (MIT), .gitignore
- /docs — Final project report (PDF), Software Requirements Specification
- /diagrams — UML diagrams (Use Case, Class, Sequence, Activity, ER Diagram)
- /design — Figma export files, UI screenshots, colour palette documentation
- /src — (Future) Source code for mobile app (React Native) and backend (Node.js)
- /tests — (Future) Unit test files and test documentation

8.3 GitHub Workflow

The team follows the GitHub Flow branching strategy:

1. All new features and tasks are developed on dedicated feature branches (e.g., feature/login-screen, feature/issue-reporting)
2. Team members submit Pull Requests (PRs) for code review before merging into the develop branch
3. The develop branch is periodically merged into main after testing and review
4. GitHub Issues are used to track tasks, bugs, and feature requests
5. Commit messages follow the Conventional Commits standard (e.g., feat: add GPS verification module)

9. Project Management

9.1 Gantt Chart

The Gantt Chart below illustrates the project timeline across 9 active development weeks (Weeks 1–11, excluding buffer and submission weeks). Blue cells indicate the duration of each task. The chart is based on the Agile sprint plan defined in Section 3.4.

Task	Start	End	W4	W5	W6	W7	W8	W9	W10	W11
Planning & Requirements	Week 4	Week 4								
System Architecture Design	Week 5	Week 5								
UML Diagrams	Week 6	Week 6								
UI/UX Prototype (Figma)	Week 7	Week								
GitHub Setup & Commits	Week 5	Week 9								
Documentation / Report	Week 7	Week 8								
Testing & Review	Week 9	Week 10								
Final Submission	Week 11	Week 11								

Figure 3: AquaGuard SL – Project Gantt Chart (Weeks 1–11)

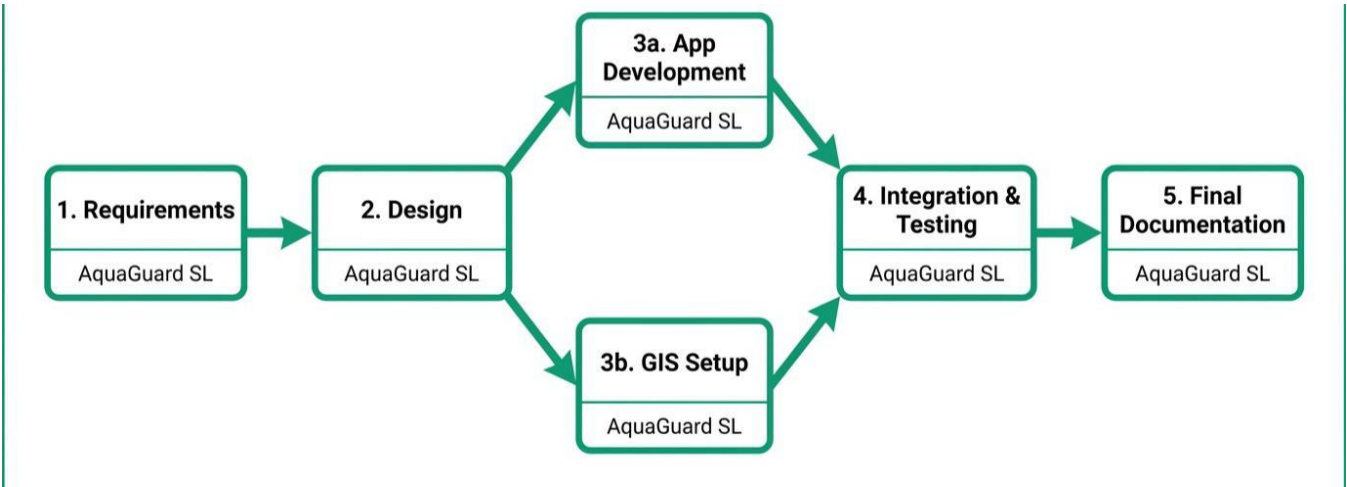
9.2 Network Diagram (Critical Path Method)

The Network Diagram uses the Critical Path Method (CPM) to identify the sequence of dependent tasks that determine the minimum project duration. The critical path is: A → B → D → G → H, totalling approximately 8 weeks of dependent work.

Node	Activity	Depends On	Duration
A	Planning & Requirements	—	2 weeks
B	System Architecture Design	A	1 week
C	UML Diagrams	A	1 weeks
D	UI/UX Prototype (Figma)	B, C	1 weeks
E	GitHub Setup & Initial Commits	B	1 week
F	Documentation / Report Writing	D, E	2 weeks
G	Testing & Review	D	1 week
H	Final Report & Submission	F, G	1 week

Figure 4: AquaGuard SL – Network Diagram (CPM)

Critical Path Analysis: The longest path through the network is A → B → D → F → H, totalling 10 weeks. This path has zero float, meaning any delay in these activities will delay the final submission. Tasks C (UML Diagrams) and E (GitHub Setup) have 1-week float and can be slightly delayed without affecting the deadline.



9.3 Risk Management

Risk	Likelihood	Impact	Mitigation Strategy
Team member unavailability	Medium	High	Cross-train all team members; redistribute tasks during sprint planning
Scope creep (feature expansion)	High	Medium	Strict backlog management; only approved features enter current sprint
Figma prototype delays	Low	High	Begin design in Week 4; daily check-ins; use component libraries
GitHub merge conflicts	Medium	Low	Feature branches; mandatory PR reviews before merging
API integration complexity	Medium	Medium	Use mock APIs during prototype phase; integrate real APIs in development phase

10. License, Design Quality & SDG Relevance

10.1 Software License

AquaGuard SL is released under the MIT License — one of the most permissive and widely used open-source software licenses. The MIT License was selected because it:

- Allows free use, modification, and distribution of the software by anyone
- Permits commercial use and integration into proprietary systems
- Requires only that the original copyright notice and license text be included in all copies
- Aligns with the Digital Public Goods (DPG) Standard, which requires open-source licensing
- Encourages adoption by NGOs, governments, and developers across West Africa

License Name	MIT License
SPDX Identifier	MIT
Copyright Holder	AquaGuard SL Development Team, 2026
Permissions	Use, copy, modify, merge, publish, distribute, sublicense, sell
Conditions	Include original copyright notice and license text in all copies
Limitations	No warranty or liability provided by copyright holders
License URL	https://opensource.org/licenses/MIT

10.2 SDG Relevance & Alignment

AquaGuard SL is designed as a Digital Public Good with direct and measurable alignment to multiple United Nations Sustainable Development Goals (SDGs). The table below outlines the primary and secondary SDG alignments:

Type	SDG Goal	Alignment & Impact
Primary SDG	SDG 6 – Clean Water and Sanitation	AquaGuard SL directly enables citizens to report water issues, supports WASH officers in issue tracking, and ensures communities have access to safe, clean water information in real-time.
Secondary SDG	SDG 3 – Good Health and Well-Being	By rapidly detecting contaminated water and alerting communities, the system helps prevent waterborne diseases such as cholera, typhoid, and dysentery.

Type	SDG Goal	Alignment & Impact
Secondary SDG	SDG 11 – Sustainable Cities and Communities	The platform supports urban and rural water infrastructure management, helping local governments maintain sustainable water supply systems.
Secondary SDG	SDG 17 – Partnerships for the Goals	AquaGuard SL bridges government WASH departments, international NGOs, health workers, and citizens in a shared digital platform for coordinated water management.

AquaGuard SL contributes to SDG 6 Target 6.1 (safe and affordable drinking water for all) and Target 6.b (support and strengthen the participation of local communities in improving water and sanitation management). The platform serves as a direct tool for community engagement in water governance — a critical requirement of the SDG 6 framework.

10.3 Digital Public Good (DPG) Compliance

AquaGuard SL meets all nine Digital Public Good Standard indicators:

- Relevance to SDGs — Directly addresses SDG 6 (Clean Water and Sanitation)
- Use of Approved Open License — MIT License (OSI-approved open-source license)
- Clear Ownership — Copyright held by AquaGuard SL Development Team, 2026
- Platform Independence — Cross-platform (iOS, Android, Web), no proprietary lock-in
- Documentation — Complete technical documentation, README, and user guides
- Mechanism for Extracting Data — RESTful API with open data export (JSON, CSV, PDF)
- Adherence to Privacy & Data Security — GDPR-aligned data practices, encrypted storage
- Adherence to Standards & Best Practices — Follows W3C accessibility, OWASP security guidelines
- Do No Harm — No collection of sensitive personal data beyond what is necessary

11. Conclusion

AquaGuard SL represents a significant step forward in addressing Sierra Leone's water and sanitation challenges through digital innovation. By creating a direct, accessible channel between communities and water management authorities, the platform has the potential to dramatically reduce the time between issue discovery and resolution, prevent waterborne disease outbreaks through early contamination alerts, and build evidence-based data infrastructure for water resource investment decisions.

This final project has demonstrated the application of core Software Engineering principles across the complete Software Development Life Cycle — from stakeholder analysis and requirements gathering, through system architecture design and UML modelling, to high-fidelity prototype development and project management. Each phase of the project has been documented to professional standards in accordance with the assessment criteria.

The Agile SDLC approach adopted for AquaGuard SL proved highly effective for this type of community-driven system, where requirements evolve with stakeholder input and iterative delivery is essential to maintaining alignment with real-world needs. The three-tier architecture provides a scalable, secure, and maintainable foundation for future development.

The prototype, designed using Figma and aligned with mobile-first accessibility principles, demonstrates a user experience that is intuitive enough for community members with minimal digital literacy, while providing the advanced analytics and coordination tools required by government agencies and NGOs.

As an open-source Digital Public Good licensed under the MIT License and aligned with UN SDG 6 (Clean Water and Sanitation), AquaGuard SL is designed not only for Sierra Leone but as a replicable model for water issue management across West Africa and beyond. The platform embodies the conviction that technology, when designed with and for communities, can be a powerful force for social change.

Future development phases will focus on: implementing the full React Native and Node.js codebase, integrating real Google Maps API and Firebase Cloud Messaging, conducting field testing with community users in Freetown, and submitting AquaGuard SL for formal recognition as a Digital Public Good through the DPG Alliance registry.

12. References

The following sources were consulted and referenced in the development of AquaGuard SL:

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